Integrated information theory → everyone hates this theory because is the most reported e complete theory of consciousness (angry that their theory is not considered as much as this). There is a lot of effort by the supporters of this theory (a lot of mathematical test and prediction to try and attempt validating this theory). Any system can be conscious, depends on the connection that you have. Measure called integrated information (Phi Φ), computed mathematically, but right now only in models, the moment you go on brain it doesn't work because is too complicated. Phi cannot be measured on humans directly, related to entropy. The problem is that based on this idea I don't need the frontal cortex in order to be conscious, even a set of neurons that reach a certain degree of information integration can be conscious for themselves (main point of disagreement). The frontal cortex is only necessary if I need to report my experience, but then to be conscious I don't need it because the posterior regions of the brain have a sufficiently complex structure to support consciousness.

They tried to validate one of the theories as better than the others and no theory won. Only one got 2 out of 3 and the other ones 1.5 out of 3. A journalist said that Giulio Tononi's theory won (information integration) and so everyone else raised up to say that it was bullshit. The effect in which scientists are against each other is not good for research.

Information is high where there is integration and communication.

Lesson 22/10/2025 (Learning and memory)

What happens in the brain when we learn and acquire information?

Learning is the process that lead to the acquisition of new knowledge or skills, memory is the product of it (acquired knowledge and skills). Learning leads to some kind of changes in the brain that allows the brain to store the information acquired. Typically we go through 3 steps while learning:

- Encoding, that is the storage of information.
- Consolidation, most of it happens during sleep.
- Retrieval, the one in which you access the memory, and you use it in behaviour.

To acquire information, we need to rely in our senses, not only the main 5 but also through the interoceptive information processing, that acquires information from the state of the body. We learn to define emotional states based on what we feel in specific conditions (swaeting, acceleration in the heartbeat, etc.).

Once that the information reaches the brain there is a sort of sensory memory, which is the most basic but also fleeting (few ms), that is the time of processing of the information that we acquire. The information stored somewhere in the brain, and this is called short-term memory, the other name is working memory. From this type of memory you can retrieve and use information, or send it in long time memory through consolidation, that continues for days, months or even years, at least few hours.

There can be also some loss of information, that can happen at all the steps, and it also may happen in all the moments in which we retrieve information from memory, and they become less accessible each time. If you learn something for an exam you see that memory weakens over time, in a point that you don't remember almost anything.

The terms short-term memory and working memory are strongly related (almost synonymous), even though the second one is more recent, introduced after realizing that when something is stored in short-term memory it's not just there but is also used by the brain, and they started to have the idea that there could have been something more similar to running memory in computers, able to put something in memory to process it and then you can remove it or store it.

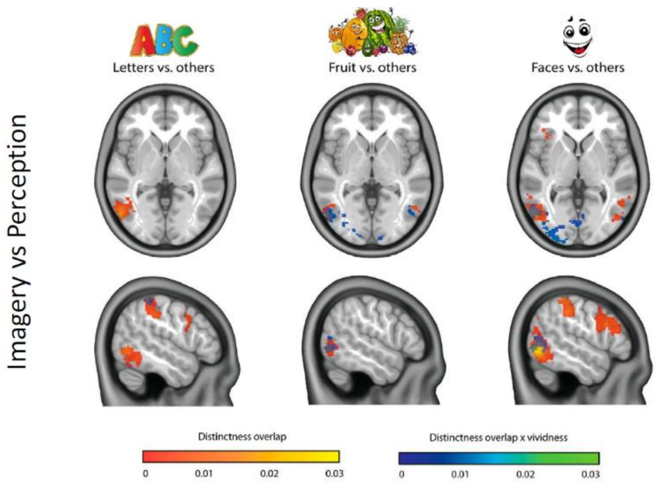
Short-term memory is the brain's temporary storage for a small amount of information that can be actively recalled for a brief period and is essential for tasks like holding a phone number or maintaining a conversation. Useful when you need to follow someone that is talking to you and maybe reconstruct the meaning of a sentence. The working memory can instead be manipulated. This type of memory has some characteristics:

- Is limited, if you try to memorize a number you can see there is a limit (4-7, found with studies).
- It has a brief duration, meaning that you typically store information from seconds to minutes, until it's lost or stored in long-term memory.
- The information is held in an active, readily available state for immediate use, contrary to long-term memory, in which we have information that does not need to be recalled in continuation and remains silent.
- It also serves as a temporary space for information before it is forgotten or transferred to long-term memory.

The encoding of memory during sleep is altered, but if something wakes you up in that moment you are able to remember the dream and encode it. The information can stay in the working memory for some time, and we can revisit and stabilize the memory of the brain or get distracted by the things we have to do and then lose it.

Memory is something that affects the brain development, usually those who become blind after the age of six have visual memory, so they are able to detect colours and remember images, while those who become blind before this age are unable to recall any visual experience. Furthermore, most of the memory you have of when you were 5-6 years old is probably false memory, something that you created based on what your parents told you or on photographs or video that you saw, but then after seven it stabilizes.

Some forms of imagery (create something from memory) can be considered also as a form of internal perception. When you do that you activate the same exact brain regions that you activate when you perceive, for example, an apple.

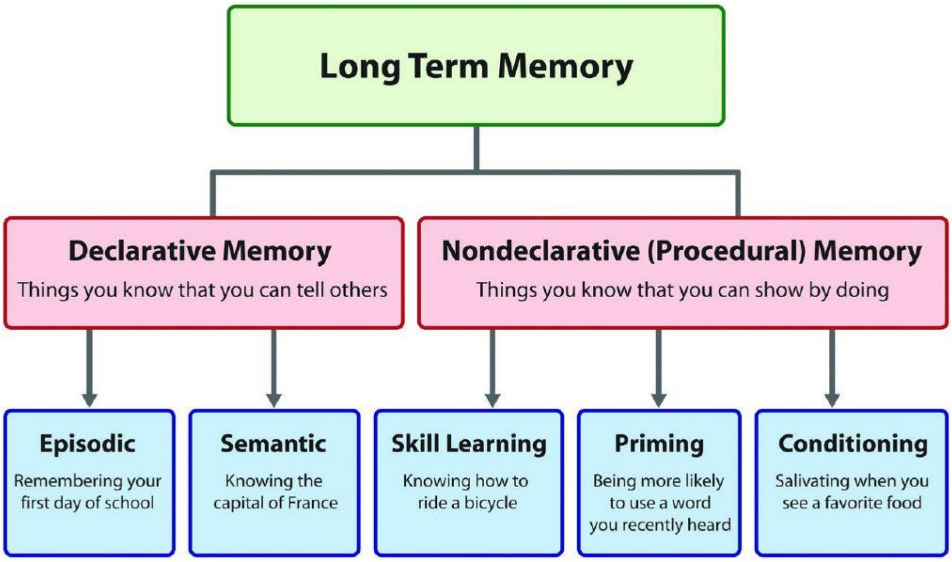


Here you can see the overlap of what you can see in the brain between the activation of the brain during a perception of a face and the imagery of a face. The region is situated in the inferior posterior part of the brain, that is specialized in perception of human faces.

What changes is that when you perceive something from the external world this happens because through the eyes you acquire information and there is a flow of information from the posterior part of the brain (occipital cortex) to the anterior part, so there is a distribution of activity across different regions. When you activate your memory (for instance of faces), the command starts from the anterior part of the brain, that retrieve the information about the face, for example of someone you know, and you have an activation in the cortex in the corresponding brain region. As long as you keep the information in mind you have activation in some frontal regions that are important to keep the information active in this top-down manner, and you have activation in the brain regions that are needed to process information.

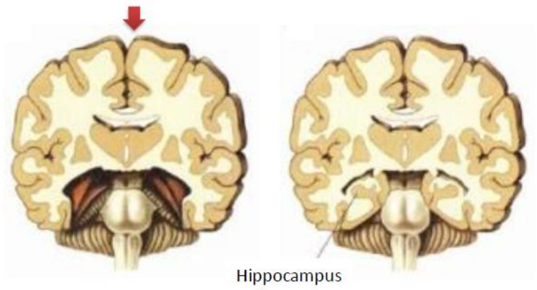
When we talk about memory we have different types of definitions. We focus on the one in which we divide memory:

- By duration. Saying that the shorter one is the sensory memory, related to the processing of the sensory information. Then we have the short-term memory and working memory (order of seconds-minutes). And finally, we have the long-term memory, that has potentially a duration of all life.
- By content. We specifically divide memory as declarative (explicit) or non-declarative (implicit). The difference is basic, meaning that implicit memory is for example learning how to ride a bike, acquiring skills in general that allows you to bike forever, and this is something that stabilize in the brain and helps you also with other activities. The declarative memory is instead the memory of facts, and it's the report of facts. These are partially dissociated in general; you can lose memory of facts but may maintain implicit memory, so they are supported by different parts of the brain.
- By process. Different phases of memory acquisition: encoding; consolidation; retrieval; reconsolidation. The last one is related to the process of retrieving the memory and stabilizing it and storing it again.



Some examples of different the different types.

If we talk about memory, we have to talk about Henry Gustav Molaison, that was known until he died as patient H.M. He had brain surgery for a seizure, and since it started from a specific region of the brain, that is the hippocampus, the doctor at the time thought they could just take out it. Removing the hippocampus (anterior-interior portion of the brain) he was not able to form new memory at all, not able to acquire any information. They even tried to give him cross-words and once completed and cancelled, the second time they presented it to him he thought that he had never seen it.



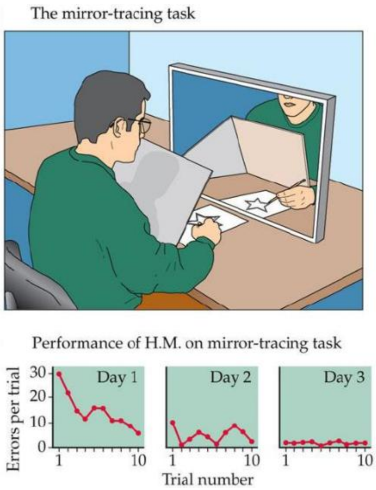
Permanent anterograde amnesia due to bitemporal resection performed in an attempt to cure epilepsy

When this person died, the brain was donated to science, and analysing it was seen that the hippocampus was not completely destroyed but most importantly they cut the connections between it and the brain, so that even the remaining parts of the hippocampus were not able to communicate. This person was able to maintain working memory, but then when distracted his memory was lost again. The take on message is that the hippocampus is crucial for the brain to acquire and store information.

The hippocampus is close to other structures, like the amygdala (important for emotional processing), emotions are important in memory. It is also close to the olfactory cortex, in fact the smell of something can activate the retrieval of memory that we have in mind, because they have a direct access to the hippocampus. Furthermore, this structure is able to receive information both from the extern and the intern of the body, for example visual information is processed in what is called two streams, the dorsal stream processes the spatial context of the information (where something is in an environment), while the ventral stream processes the identity (the what of the object). There are two path that elaborate mostly different aspects of the information from the extern, then transferred to the hippocampus, but also internal information, for example from the amygdala, that processes the emotional context of what is acquired.

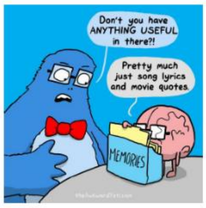
The idea about the context and the content is important, because short and long-term memories are both limited, the information that we can keep in the brain is not infinite. This become very clear when is asked to remember face from old famous people, because we tend to forget faces to create store for new ones.

H.M. was then asked to do another task, he had to draw in a paper that he couldn't see if not as the image on the mirror. You have to learn how to draw in a mirrored way, and measuring the number of errors he did they found that there was an improvement in the performance during the day that was also maintained the following days, but then when he was asked if he remembered that he had already did the task he would have said no. So there was an improvement in precdural memory, his skills. The two types of memory are separated, cerebellum for example is crucial for motor skills and also other structures, so that the motor cortex exchange information with them making the patient able to improve motor abilities. Motor memory doesn't rely on the hippocampus.



Preserved working memory and intellectual abilities.

Maintained ability to acquire new visuo-motor skills (procedural/implicit memory).



The patient lost the ability of creating new memory, but also other memories from few years before the incident. Some memories were actually not yet transferred in the long-term storage, they were still being consolidated.

Encoding

The memory is somehow stored in the brain, so that it changes in order to incorporate this type of information. This is an engram (not sure of what it is), a physical trace of memory in the brain, a neural correlates of memory, what happen in the brain corresponding to the storage of memory. This has 4 characteristics:

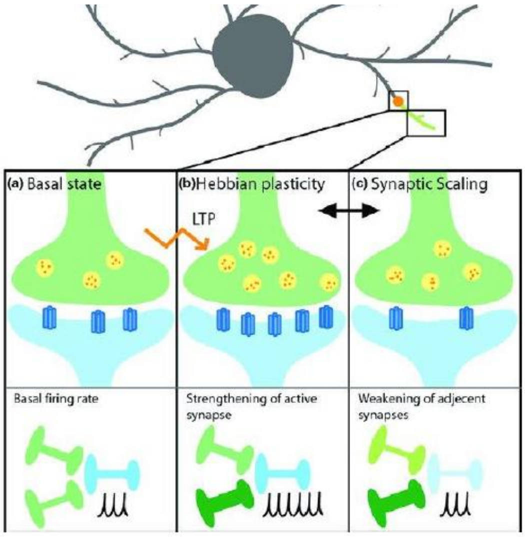
- An engram is a persistent change in the brain that results from a specific experience or event, that last forever;
- The engram should have the potential to influence the behaviour, brain change → change in behaviour (you put a finger in fire you don't do it anymore). This comes from the interaction with retrieval cue, you recall what you did and felt in life because the cue has this power.
- The content of the engram reflects what happens at the encoding and is reactivated during retrieval. Easy to understand if you think about the concept of imagery, those regions are also involved in the retrieval process.
- The engram is stored in a sort of silent way, unless you reactivate the engram voluntarily or there is a cue that does it, the engram is there and if not used it's just there.

An engram provides the necessary physical conditions for a memory to emerge.

This trace in practice may be related to how neurons are connected, and the strengthening or weakening of connections (simplification). But there are other cells that for a long time were only considered as supporters of the neurons, just as astrocytes, that were discovered to do a lot of work in the brain, but this is just for the moment an idea. The problem is that neurons because of their electrical activity produce something that can be actually measured, but astrocytes do not produce a very clear signal (at least in humans). With f-MRI maybe we are also measuring their activity in the change of blood flow, caused by their request of oxygen, but from this type of technique we are not able to see which structure is demanding more blood flow.

The engram is been hypostasized to be mostly a change in the strength of connection between neurons, so when we get new information neurons will get the brain activated and this activation will generate some kind of interaction between neurons, with a strengthening of their connection. This is called Hebbian plasticity, saying that neurons that fire together wire together. There are group of neurons that work together for the same task, for example when seeing an apple in the brain the neurons that activate for red objects, the ones for natural objects, will start to fire together to represent the apple in the mind, so those neurons filed together will start to connect among themselves strengthening it. If I see an apple only once in my life that image doesn't remain in the brain being useless, but if I see a lot of them this information become important for the brain and is consolidated in it.

This process of strengthening is called long-term potentiation, and it is a persistent strengthening of synapses based on recent patterns of activity that produce a long-lasting increase in signal transmission between two neurons. The opposite of LTP is long-term depression (LTD), which produces a long-lasting decrease in synaptic strength.

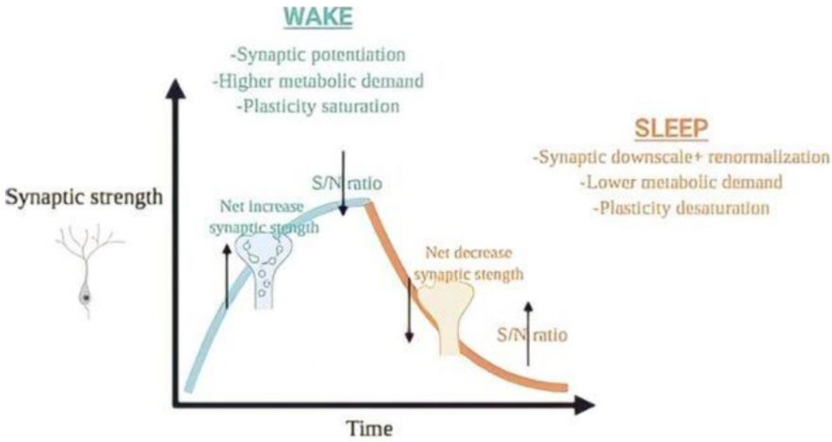


When strengthening it can happen that their synaptic button may become larger, so that I am able to release more chemicals and be more efficient. But it may also represent a growth in the number of chemicals stored in the pre-synaptic button, so that I can release more of them and be more efficient. But also, an increase in the number of receptors on the post-synaptic button. This type of event obviously requires time, even minutes, hour or even days.

The brain tends to select what is important for it, based on different things. Firstly, the number of times you see the element. Synaptic plasticity: ability of the brain to strengthen, weaken, destroy and create neural synapses as the basis for learning. The effect of a learning experience depends on the content of such an experience. Reactions that are favoured will be reinforced and those that are deemed unfavourable will be weakened.

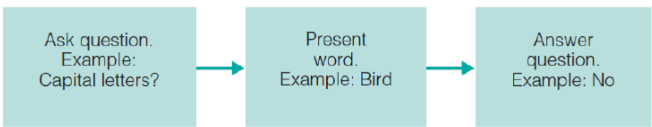
- Short term: strengthening or weakening of a connection by modifying the preexisting proteins leading to a modification in synapse connection strength.
- Long term: new connections may form or the number of synapses at a connection may be increased or reduced.

During the day is mostly activated by all the situations that happen around us, and many of our synapsis are potentiated. This leads to a saturation of synapsis and more energy needed by the brain resulting in a problem for it, since it is not sustainable on a long-term. For this reasons a theory has been proposed, the synaptic homeostasis hypothesis (SHY), that tells that basically this type of increase is compensated during sleep, and we have some proof of this that in some regions the synapsis strength decrease (downscale), but this downscale doesn't happen in all the synapsis. Some synapsis that are bigger and more important are not downscaled while others less important are weakened, leading to an increase of what is called Signal to Noise ratio. This mechanism is what make us able to learn new things every day. Resuming synaptic potentiation during waking is connected to a homeostatic increase in slow-wave activity (SWA) during sleep, during which the systematic normalization of synaptic strength (synaptic downscaling) takes place. Larger and most potentiated synapses may remain unscaled. Weak connections may be eliminated, whereas the relative strength of the remaining connections is preserved.

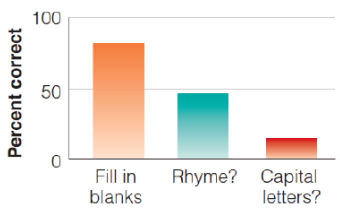


This type of studies were conducted on animals, because in humans the evidence is a bit more indirect, meaning that they relied for example in the potential that you can induce with TMS, so that if you have many neurons strongly connected and you sent a pulse, you get a response that propagates very efficiently, validating the theory of potentiation. If you do the same thing during sleep, you get a much smaller reaction, the brain is less efficient, and this may happen because the neurons are not strongly connected. Unfortunately, at the moment, we are not sure it really happens in all brain regions in the same way, for example in the motor cortex the evidence is clear, while in other, e.g. visual cortex this effect was not observed. A theory is that the visual cortex doesn't have to change that much during the day because our perception of the world it's the same, while the motor cortex has to do a lot of things.

Depending on if you want to store information, the depth at which you elaborate information can affect how much you are really able to store information.



(a)



(b)

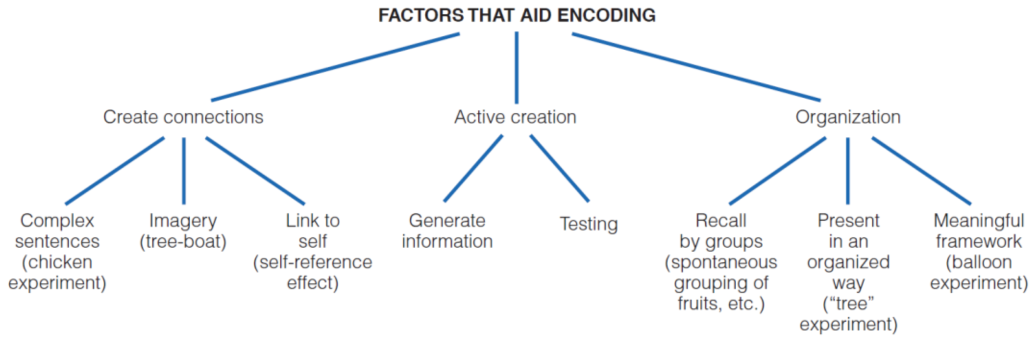
1. **Shallow** processing: A question about **physical features**
Question: Is the word printed in capital letters?
Word: bird
2. **Deeper** processing: A question about **rhyming**
Question: Does the word rhyme with train?
Word: pain
3. **Deepest** processing: A **fill-in-the-blanks** question
Question: Does the word fit into the sentence "He saw a on the street"?
Word: car

Experiment: participants had to memorize the larger number of words possible and then they had to rely on these different levels of processing. What came up was that if you have the deepest level of processing (reflection on the meaning of the word) you are much better at remembering the words compared to a more superficial listing of the word. So, the level of processing affects the capability of storage of the word.

Another study performed considered the effect of imagery. If I associate imagery, can I potentiate the storage of information? It was found that associating a visual corresponding to the word, you are much better at memorize the word. Imagery potentiates the storage of information in the brain.

The other aspect is the relevance for the person, same task of before (the level of processing) adding another condition in which they asked the patient to try and focus on the relevance of the word for him, imaging in some way a connection between it and himself. It was found that it potentiates the storage of information. For example, when you study something that you found interesting.

The other thing is the structure of what you have to learn. If you have a list to memorize and you try and learn in a total unstructured way, it's much more difficult than trying to separate the words in clusters and find connection between the elements.



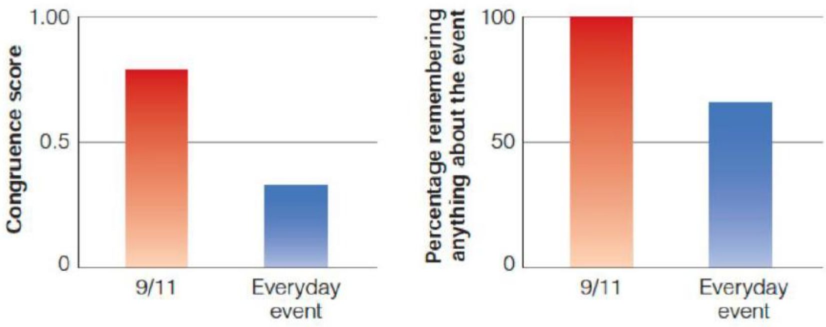
One thing that is also more important is the emotional connection. If something that has an emotional charge for the patient is shown to him this will potentiate the storage of information. If pictures or words that are associated with an emotional content are presented, they are much easier to remember. And this is why you have post-traumatic stress disorders, anxiety disorder that comes from a traumatic event. The strong emotional involvement makes the memories associated with it basically stamped in the brain in such a way that you can even revive the event vivid in the mind. You are in a condition of strong anxiety because this event can intrude spontaneously in your memory or caused by a cue. PTSD is the only case in which you dream about an event as it really had happened, and not an elaboration. One way to treat PTSD is to revive the memories, because when you reactivate a memory you can weaken it, and try to change what happened in the event.

We know that noradrenalin is crucial in the formation of new memories, and in case of strong emotional involvement we have a large release of noradrenaline and so this is really imprinted in the brain. Ketamine is not the best medicinal to treat this disease because it creates hallucinations.

Everyone that went through the 9/11 event remember what they were doing at that time. These types of memories are called flashbulb memories: vivid, detailed, and enduring memories of the circumstances in which someone first learned about a surprising, emotionally arousing, or consequential event. They have different key features:

- Vividness: people often recall memories with a strong sense of detail (where they were, who they were with, what they were doing).

- Emotional intensity: typically tied to highly emotional or shocking events.
- Confidence: individuals are usually very confident about the accuracy of these memories.
- Content: often includes contextual details (location, time of the day, ongoing activity, emotional reactions).



However, these memories are often not different from other memories, even if we have this vividness of the memory, we can always have an alteration of the memory. Example: for the challenger space shuttle explosion, in a time frame of two years a woman changed version of the facts reported because her memory changed. The memory is partially different, there is a lack of the first portion, maybe she deleted it.

Exactly for all the other memories, even this type goes down over time, but one thing that is interesting is that this memory is reported by the person with a great confidence. And this is an effect of the emotional involvement. Emotions are crucial component, whether something will be stamped in long-term memory and also affects the confidence that is in the person talking about it.

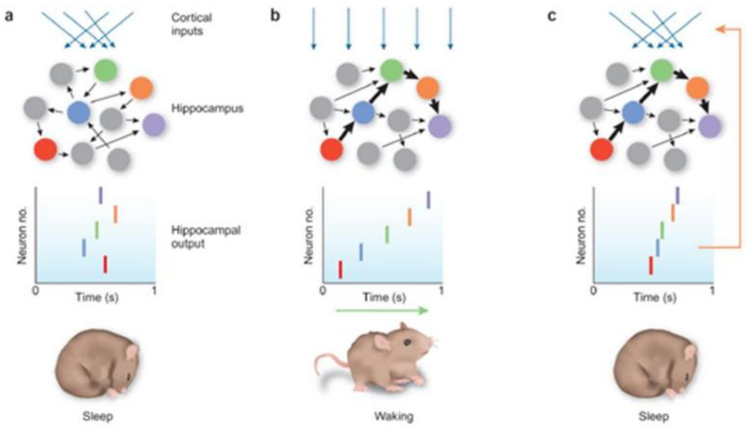
The memories can change over-time but at least at the beginning they are consolidated, and this process of consolidation happens every time we keep our brain at rest (having the time to ri-elaborate stuff), especially during sleep. But how consolidation works?

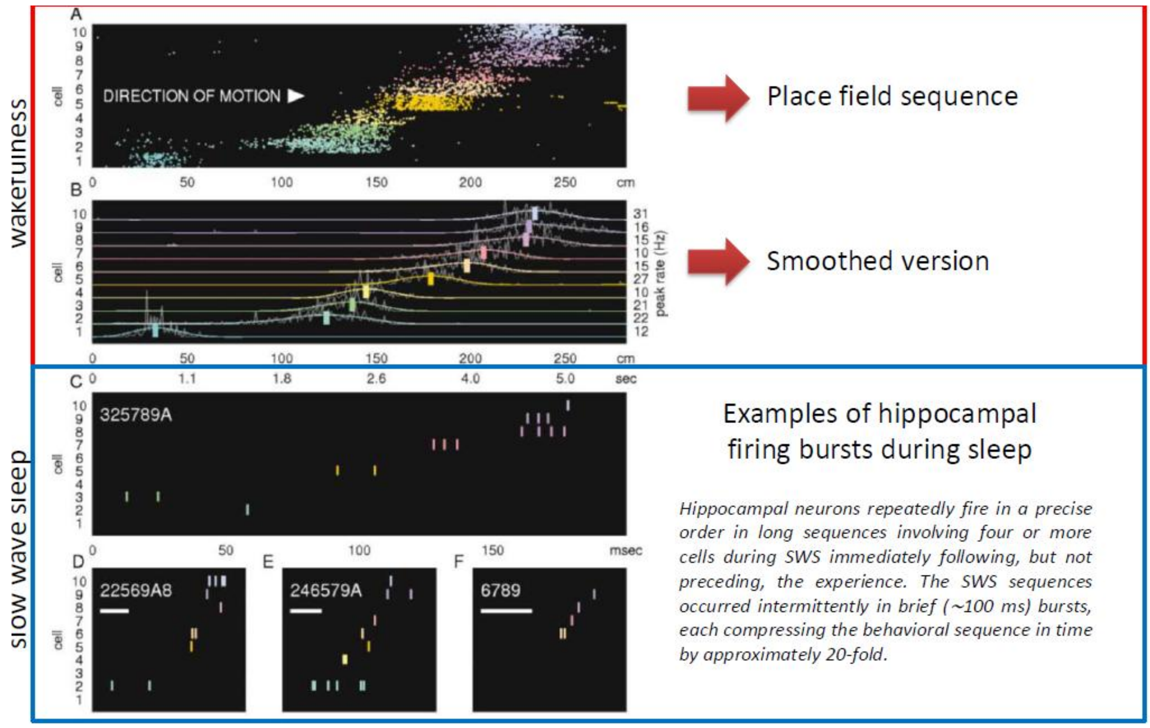
- **Synaptic consolidation** (initial consolidation): Is achieved faster than systems consolidation (minutes-hours). Depends on Long-term potentiation (LTP). Writing in the brain, with the formation of the engram and the adaption of the connections between the neuros.
- **Systems consolidation** (how the memory is organized within the brain): It is a reorganization process in which memories from the hippocampal region, where memories are first encoded, are moved to the neo-cortex in a more permanent form of storage. Takes days to months.

Hippocampus is crucial for memories to be stored, but not for old memories, that must be stored somewhere else. This is what the consolidation process does, it stabilizes the memory and transfers in part the site of the memory. As we said there is a link between the neurons in the cortex and some neurons in the hippocampus, that basically are at least at the beginning controlling the memories, with no hippocampus this process will never become stable (no consolidation process). Then when I go to sleep the connections between the cortical and hippocampal modules weaken and at the very end (even years), the memory becomes independent from the hippocampus. Obviously we are talking about memories that depend on the hippocampus, not procedural memories that have different processes of consolidation.

This is called Ribot's Law: a principle of retrograde amnesia, proposed by French psychologist Théodule Ribot in 1881, which states that recent memories are more vulnerable to loss than older, more remote memories (classical example is dementia, first lose recent memories and nothing of the day, no hippocampal mechanism). This temporal gradient in memory loss suggests that memories undergo a consolidation process over time, becoming more stable and resistant to disease or damage as they age.

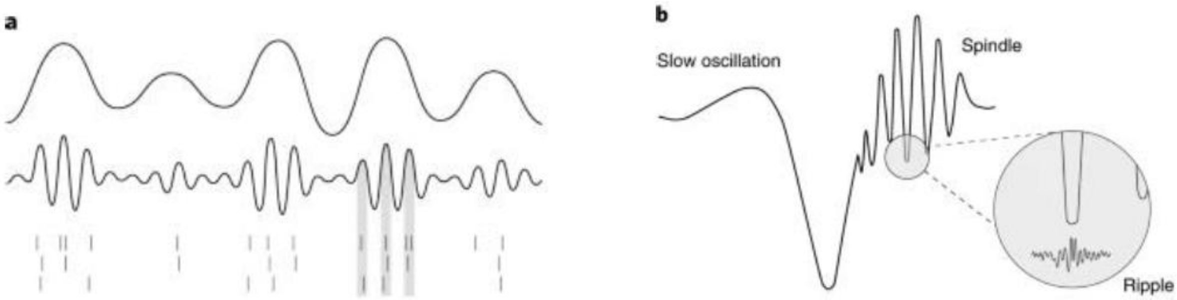
What happens during sleep? Experiment: you create a labyrinth and implant electrodes in the hippocampus, where there are some cells that record and respond specifically to the spatial position and the direction of the animal, so when the animal starts to move there are some neurons the respond when it goes straight, left or right. You can give colours to these neurons and track the path of the mouse inside the labyrinth. Since the mouse learn the path, it is found that while sleep there is an activation of these neurons in the hippocampus in the exact same sequence, repeated several times (replay). Sometimes it is also reactivated backwards, several times.





We found evidence of a similar process also in humans, more indirect. During sleep when involved in this memory task there is an activation especially in the hippocampus, trying to replay the event. There are studies that suggest that these processes can also last for several days, and it's for sure something that needs some time to happen. Replay events in the V1 and HP were coordinated to reflect the same experience. These results imply simultaneous reactivation of coherent memory traces in the cortex and hippocampus during sleep that may contribute to or reflect the result of the memory consolidation process.

During sleep there is a crazy association of different oscillation with different parts of the brain, so basically in the hippocampus there are ripples, very fast oscillation, that reflect this sequence of reactivation, associated with spindles (oscillation at 40 Hz, generated by the thalamus and projected to the cortex) and slow wave at the cortical level. At the cortical level you have the slow wave, that at the end has the spindle, and during the spindle you have the ripple in the hippocampus. You have a coordination of events across the cortex, hippocampus and the thalamus that allows the network that has been activated during learning, to be reactivated e reprocess the memories, these sequence of events weakens the connection between the hippocampus and the cortex and strengthens the connection at the cortical level. This is something that happens mostly during sleep.

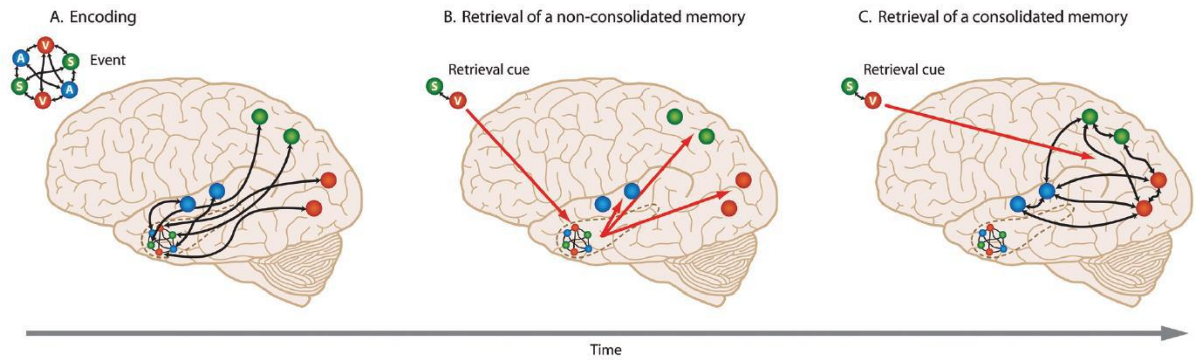


The relationship with dreams is not clear at all, but it's clear that dreams have something to do with these mechanisms of reprocessing. Right now, the only access that we have is the replay one, and in humans it is difficult to see the coordination in the hippocampus.

The retrieval is indeed the act of accessing information from memory. Retrieval can take the form of recall, in which one searches memory for a specific piece of information and is able to produce it. Retrieval is also involved in recognition, as when a person judges that they have seen or encountered something before. Retrieving information from memory is assumed to (partly) reactivate contextual features of the original experience, owing to the connections forged between the information and the surrounding context at the time of encoding. Similarly, being in an environment that shares features (cues) with the original encoding context can also facilitate retrieval of the information, this may need to mind wondering (memory starts to explore spontaneously the memories associated with something). Spontaneous retrieval is involuntary, triggered by cues with little effort, whereas voluntary retrieval is a deliberate, goal-directed process to recall past events. Spontaneous memories often arise unexpectedly, like from Proust's Madeleine story (there was a reactivation of a memory from a smell), and may feel more meaningful. Voluntary retrieval involves an intention to find a specific memory, often requiring conscious effort.

If I want to retrieve the memory when it is still not fully consolidated, B state,(from the cortex and the hippocampus) I must rely on the HC, that drives the retrieval of information. When I am in the

C state the memory is stored in the cortex, so the cue acts directly on the cortex, and that's why it is more stable.

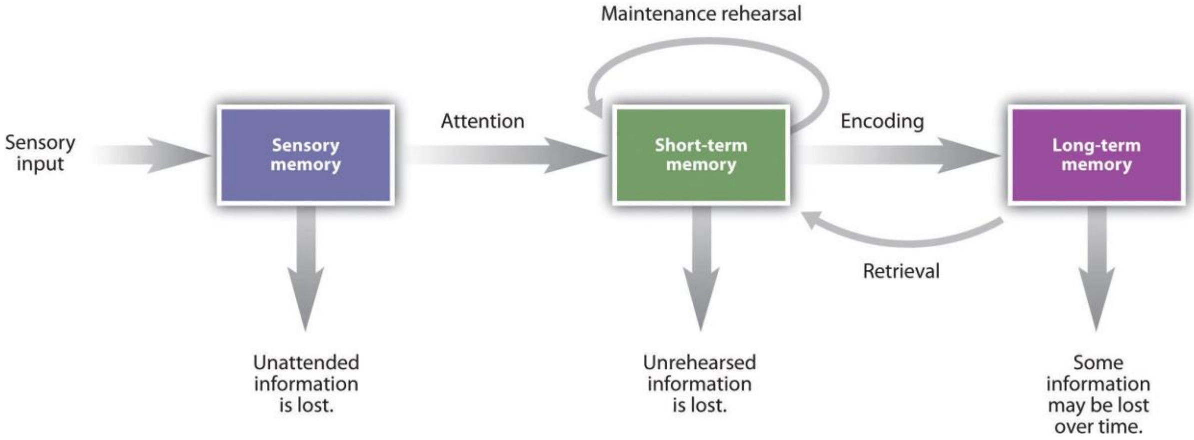


The memory can be influenced by external information to a process of competition. If I give you some images to study and you try to memorize them and then I ask you to reactivate this memory (retrieval) and in the same time I also show you other pictures, this process of retrieval will make you mix old and new images, but this doesn't happen if I don't ask you to do retrieval. Mixing is also more likely to happen if the images are acquired in the same context, that is a key element. The process of retrieval makes your memory more vulnerable to manipulation that could be voluntary or not. You should try to avoid conditions that can be an interference. Vulnerability can also be a positive thing, like in the case of traumatic memories when you try to get access to them and re-write them into something more acceptable, or in cases in which you want to improve your capability to memorize creating associations between the elements.

This applies to declarative memories, memories that are related to practice and emotional ones are more stable. Not all memories are vulnerable in the same ways; it depends on how strong the encoding and consolidation process is. Enhancing memory via retrieval. The act of accurately retrieving information at one point in time increases the likelihood that the information will be accurately retrieved at a later point in time. This phenomenon is referred to as the testing effect — because tests of memory require retrieval — or the retrieval practice effect. The retrieval practice effect is even more pronounced if participants in the retrieval condition are given feedback on their failed retrieval attempts.

If you want to make someone regain a movement, you can make them imagine the movement and this can help the process of training.

Information affected	Positive effects	Negative effects
Retrieved information	With accurate retrieval (recall or recognition), later retention is improved	The act of retrieval opens a window for possible changes in memory from both external and internal influences
Related information	In some circumstances, the act of retrieval can lead to retrieval of related information	The act of retrieval can inhibit recall of related information under other conditions
To-be-encoded information	The act of retrieval can facilitate encoding of additional information	The act of retrieval can facilitate encoding of misinformation



The brain at rest

We talk about how the brain works at rest because it has several implications for the study of the brain. Typically, we have a contrast in which we have a period of rest and a period in which we do a specific task, so we expect the regions that react to the specific stimulation to show a change in activity and then we compare the activity that we observe during the task to the images of brain at rest. So, the regions that show a response during stimulation are activated during it, involved in the task, and what we consider are the temporal and spatial correlates of the activation. What we were able to see is that while doing a task there are also regions that have a negative response, some parts get deactivated, and it may mean that during rest there are parts of the brain strongly activated. This led to the creation of one of the most widely used file dime in research with f-MRI that is called resting state, the most common type of data (simple way to detect it).

The idea at the very beginning was to collect a lot of data and see what would change in the brain for a pathological condition, and maybe even learn how to classify brains. It didn't really work out, because alterations in the brain can be found but are never specific in practice, we can have the same alterations in many pathological conditions.

There are two types of resting state, eyes open or eyes closed. The first one is typically preferred because you may fall asleep with eyes closed. The common view was that the resting state was a condition with minimal brain activity, but now the view is changed, and has been found that the increment in brain activity because of a task, it's very small (5% change in metabolic request), so even at rest the brain has a large energy consumption.